INTERIM MILITARY SPECIFICATION

LADDER, DEBARKATION; FIBER ROPE

1. SCOPE

1.1 Scope. - This specification covers fiber rope debarkation ladders of the square mesh type for use by personnel to embark or disembark over the side of a ship.

1.2 Classification. - Debarkation ladders shall be of the following grades, as specified (see 6.1):

Grade A - Manila.
Grade B - Sisal.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
T-C-621 - Cordage, Hemp, Tarred.
FF-T-276 - Thimbles, Rope.

MILITARY
MIL-C-15159 - Compound, Preservative; Fire-, Water-, Mildew-, and Weather-Resistant (for Canvas).
MIL-R-16060 - Rope, Manila; and Substitutes.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-129 - Marking of Shipments.

DRAWINGS

BUREAU OF SHIPS
S1604-939997 - Debarkation Ladders.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - The material shall be of the quality best suited for the purpose intended.

3.1.1 Fiber rope. - The rope used for grade A ladders shall conform to type M and the rope used for grade B ladders shall conform to type S of Specification MIL-R-16060, except that the specified maximum turn may be increased up to, but not exceeding 5 percent, to provide a suitable lay. Seizing stuff shall conform to type B of Specification T-C-621.

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3.2 Construction. -

3.2.1 General. - Debarkation ladders shall be furnished without wooden spreaders and spacers, shackles, and handling lines (see 6.2).

3.2.2 Size of ladders. - Debarkation ladders shall be manufactured in one width of 12 feet and lengths from 10 to 55 feet, varying in 5 foot increments, as specified (see 6.1).

3.2.3 Size of ropes and seizing. - The size of ropes and seizing used in the construction of debarkation ladders shall be as shown in table I.

Table I - Size of rope and seizing.

	Grade A - manila	Grade B - sisal
Net line	3 inch (circumference)	3-1/2 inch (circumference)
Bolt line	3-1/2 inch (circumference)	4 inch (circumference)
Seizing	6 yarn	6 yarn

3.2.4 Size of meshes. - The width of the meshes shall be not more than 11 inches on centers as shown on Drawing S1604-939997.

3.2.5 Construction of meshes. - The construction of meshes shall conform to typical ladders shown on Drawing S1604-939997. The meshes shall be formed of net lines which shall be fastened at the sides and bottom to the bolt lines. Starting at the top, the net lines shall form a crisscross pattern of interlocking meshes such that at each intersection the strands of the net line entering an intersection vertically pass under one strand of the other net line and in turn the second passes under one strand of the first, each net line making a 90 degree change of direction, as shown on the enlarged detail of an intersection on the referenced drawing and as shown on figure 1. Where the net lines terminate at the bolt line, a double tuck shall be made, as shown on the enlarged detail on the drawing, and the ends spliced into their own part with at least three full tucks, with the strands seized and trimmed. At the top, 8 beckets shall be formed for the 10, 15, 20, 25 and 30 feet ladders; and 14 beckets for the 35, 40, 45, 50 and 55 foot ladders, using type II thimbles conforming to Specification FF-T-276. Thimbles shall be of the following sizes:

Rope size Inches	Thimble size (Inches)
3	1
3-1/2	1-1/8
4	1-3/8

3.3 Fire-retardant treatment. - The debarkation ladders shall be treated with preservative compound in accordance with Specification MIL-C-15159. The preservative compound may be applied by dipping the completed ladder or by brushing or spraying. If either of the latter two methods is used, the material shall be applied to both sides of the ladder.

3.4 Workmanship. - Debarkation ladders shall be manufactured in accordance with the best commercial practice and consistent with the best rigging practice. The workmanship shall be first class in every respect and the finished ladders shall be free from any defects which may effect the serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

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4.2 Sampling for lot acceptance. -

4.2.1 Inspection lot. - For purposes of sampling, a lot shall consist of not more than 500 ladders of the same grade and length, submitted for inspection at one time.

4.2.2 Sampling for inspection. - A random sample of ladders shall be selected by the Government inspector in accordance with table II from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on 4.3.

Table II - Sampling for inspection
AQL (approx.) = 1.5 percent defective.

Number of ladders in inspection lot	Number of ladders in sample	Acceptance number (defectives)	Rejection number (defectives)
5 and under	All	-	-
6 to 8	5	0	1
9 to 15	7	0	1
16 to 25	10	0	1
26 to 40	15	0	1
41 to 65	20	1	2
66 to 110	25	1	2
111 to 300	30	2	3
301 to 500	35	2	3

4.3 Inspection. - Each of the sample ladders selected in accordance with 4.2.2 shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any sample ladder found to be nonconforming shall be rejected and if the number of defective ladders in the sample exceeds the acceptance number, the entire lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests, provided the contractor has removed (or reworked) all nonconforming ladders.

4.4 Additional inspection. - Where other specifications form a part of this specification, sampling, inspection, and tests shall be conducted as required in the referenced specifications.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing (for domestic or overseas shipment). - Unless otherwise specified in the contract or order, debarkation ladders shall be individually bundled and securely tied with fiber twine or rope to withstand handling in shipment and storage.

5.2 Marking. - In addition to any special marking required in the contract or order, bundles shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of ladders required.
(c) Grade required (see 1.2).
(d) Length required (see 3.2.2).

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6.2 Spacers, spreaders, anchor shackles, and handling and lashing line are to be supplied by the ship's force in accordance with Drawing S1604-939997.

6.3 THE ILLUSTRATIONS HEREIN ARE DESCRIPTIVE AND NOT RESTRICTIVE and are not intended to preclude the purchase of debarkation ladders otherwise conforming to this specification.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.



Figure 1. - Details of intersection.

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